

PAPER_745: Crab Nebula MUGE — Expanding Supernova Remnant with Pulsar Wind

Author: Daniel T. Murphy

Framework: Universal Quantum Field Superconductive Framework (UQFF)

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Abstract

The Crab Nebula (M1) is the remnant of SN 1054, powered by a central pulsar (PSR B0531+21) at the highest rotational energy loss in the known neutron star population. Its expanding ejecta ($r(t) = r_0 + v_r \cdot t$) creates a time-dependent gravitational environment unlike any static system. This paper derives the Crab Nebula MUGE incorporating: the expanding radius $r(t)$, the pulsar wind term F_{wind} , and the nebular magnetic energy M_{mag} . The UQFF framework captures both the mechanical expansion and the magnetospheric energy injection that power the synchrotron nebula.

1. Introduction

The Crab Nebula is a pulsar wind nebula (PWN) — one of the brightest objects in X-rays and gamma-rays. The central pulsar ($P = 33$ ms, $\dot{P} = 4.2 \times 10^{-13}$ s/s) injects $\sim 5 \times 10^{31}$ W of spin-down power into the nebulae through a relativistic wind. The expanding ejecta shell moves at $v_r \sim 1500$ km/s, with angular size growing observably over human timescales. This expanding geometry requires $r(t)$ to appear explicitly in the MUGE denominator, making the Crab the canonical example of a time-dependent gravitational radius.

Crab Nebula parameters: - Distance = 2.0 kpc - Age 970 yr (SN 1054)
- r_0 (initial remnant radius) 3×10^1 m - $r_{current}$ 5.6 ly = 5.3×10^1 m
- $v_r = 1500$ km/s = 1.5×10^4 m/s - M_{ejecta} 4–5 $M_\odot$ - B_{nebula} 10 T (filamentary magnetic field) - $L_{pulsar} = 5 \times 10^{31}$ W (spin-down luminosity) - $P_{spin} = 33$ ms

2. Crab Nebula MUGE

$$\begin{aligned} g_{Crab}(r,t) = & (G \cdot M) / r(t)^2 \cdot (1 + H(z) \cdot t) \cdot (1 - B(t) / B_{crit}) \\ & + (U_{g1} + U_{g2} + U_{g3} + U_{g4}) \\ & + U_i \\ & + (\Lambda \cdot c^2 / 3) \end{aligned}$$

$$\begin{aligned}
& + (\hbar/\sqrt{\Delta x \cdot \Delta p}) \cdot (\cdot H \cdot dV) \cdot (2/t_{\text{Hubble}}) \\
& + \text{_ejecta} \cdot V \cdot g \\
& + F_{\text{wind}} \\
& + M_{\text{mag}}
\end{aligned}
\begin{array}{l}
\text{[pulsar wind -- NEW]} \\
\text{[magnetic energy -- NEW]}
\end{array}$$

3. Time-Dependent Radius $r(t)$

The expanding radius is the defining feature of the Crab MUGE:

$$r(t) = r_0 + v_r \cdot t$$

$$\begin{aligned}
r_0 &= \text{initial radius at SN explosion (m)} \\
v_r &= \text{expansion velocity} = 1.5 \times 10^4 \text{ m/s} \\
t &= \text{time since SN 1054 (s)}
\end{aligned}$$

For $t = 970 \text{ yr} = 3.06 \times 10^7 \text{ s}$:

$$\begin{aligned}
r(970 \text{ yr}) &= 3 \times 10^{16} + 1.5 \times 10^4 \times 3.06 \times 10^7 \\
r(970 \text{ yr}) &= 4.6 \times 10^{16} \text{ m} = 4.9 \text{ ly}
\end{aligned}$$

Gravitational deceleration:

$$\begin{aligned}
g_{\text{grav}}(t) &= G \cdot M_{\text{ejecta}} / r(t)^2 \\
g_{\text{grav}}(970 \text{ yr}) &= 6.674 \times 10^{-11} \times (4 \times 4 \times 1.989 \times 10^3) / (4.6 \times 10^{16})^2 \\
g_{\text{grav}} &= 5 \times 10^{-22} \text{ m/s}^2 \quad (\text{extremely weak - expansion dominates})
\end{aligned}$$

4. F_{wind} — Pulsar Wind Term

The pulsar wind carries relativistic particles outward, creating an inward reaction force equivalent to:

$$F_{\text{wind}} = L_{\text{pulsar}} / (4 \cdot r(t)^2 \cdot c \cdot M_{\text{ejecta}})$$

$$\begin{aligned}
L_{\text{pulsar}} &= 5 \times 10^{31} \text{ W} \\
r(t) &= \text{current nebula radius} \\
c &= 3 \times 10^8 \text{ m/s}
\end{aligned}$$

$$\begin{aligned}
F_{\text{wind}} &= 5 \times 10^{31} / (4 \cdot (4.6 \times 10^{16})^2 \cdot 3 \times 10^8 \cdot 4 \times 1.989 \times 10^3) \\
F_{\text{wind}} &= 4.8 \times 10^{-3} \text{ m/s}^2 \quad (\text{small but physically significant over 970 yr})
\end{aligned}$$

Cumulative momentum injection over nebula lifetime:

$$\Delta p_{\text{wind}} = L_{\text{pulsar}} \cdot t / c = 5 \times 10^{31} \times 3 \times 10^7 / 3 \times 10^8 = 5 \times 10^{33} \text{ kg} \cdot \text{m/s}$$

5. M_mag — Magnetic Energy Term

The nebular magnetic field stores and dissipates energy, affecting dynamics:

$$M_{\text{mag}} = B^2 / (2 \cdot \rho \cdot r(t) \cdot v_{\text{ejecta}}) \quad [\text{magnetic pressure divided by density}]$$

$$\begin{aligned} B &= 10 \text{ T (filamentary field)} \\ \rho &= 4 \times 10^{-14} \text{ H/m} \\ v_{\text{ejecta}} &= 10^{-2} \text{ kg/m}^3 \text{ (current)} \\ M_{\text{mag}} &= (10^{-14})^2 / (2 \cdot 4 \times 10^{-14} \cdot 4.6 \times 10^{17} \cdot 10^{-2}) \\ M_{\text{mag}} &= 2.7 \times 10^{-1} \text{ m/s}^2 \end{aligned}$$

This is the dominant non-Newtonian term for the Crab, controlling the synchrotron brightness evolution.

6. UQFF Gravity Components

$$\begin{aligned} \text{U_g1: Pulsar magnetic dipole} \\ \rho_{\text{dipole_PSR}} &= 3.8 \times 10^3 \text{ J/T (from P, P)} \\ B_{\text{PSR}} &= 3.8 \times 10 \text{ T (surface field)} \\ \text{U_g1 contributes at pulsar vicinity only} \\ \\ \text{U_g2: Superconductive aether field threading nebula} \\ B_{\text{super}} &\text{ aligned with pulsar rotation axis} \\ \text{U_g2} &= 10 \text{ J/m}^3 \text{ (strong near pulsar wind shock)} \\ \\ \text{U_g3: External galactic field at 2 kpc} \\ \text{U_g3} &= G \cdot M_{\text{gal}} / r_{\text{gal}}^2 \\ \\ \text{U_g4: Galactic center contribution} \\ k_4 \cdot \rho_{\text{vac}} \cdot [SCm] \cdot (M_{\text{bh}} / d_{\text{g}}) \cdot e^{(-t)} \cdot \cos(\omega \cdot t_{\text{n}}) \end{aligned}$$

7. Temporal Evolution Summary

Time (yr)	r(t) (ly)	g_grav (m/s ²)	M_mag (m/s ²)
0	0.3	2×10^{-1}	10^{-1}
100	1.6	5×10^{-21}	10^{-1}
970	4.9	5×10^{-22}	2.7×10^{-1}
10,000	~50	$\sim 5 \times 10^{-2}$	$\sim 10^{-1}$

The magnetic term M_mag remains important relative to g_grav throughout nebula evolution.

8. Conclusion

The Crab Nebula MUGE demonstrates that time-dependent radius $r(t)$ is essential for accurately modeling supernova remnant gravity. The F_{wind} pulsar injection and M_{mag} magnetic energy terms together dominate over classical Newtonian gravity within the nebula. The UQFF successfully models the transition from SN ejecta to mature PWN through its modular environmental forcing framework.

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